# Configure host access

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# Configure host access

## Create host automatically

You can allow the Host Context Agent (HCA) to automatically detect the hosts, and then verify that the information is correct. Creating a host is one of the steps required to let the storage array know which hosts are attached to it and to allow I/O access to the volumes.

#### Before you begin

The Host Context Agent (HCA) is installed and running on every host connected to the storage array. Hosts with the HCA installed and connected to the storage array are created automatically. To install the HCA, install SANtricity Storage Manager on the host and select the Host option. The HCA is not available on all supported operating systems. If it is not available, you must create the host manually.

#### Steps

1. Select **Storage › Hosts**.

   The table lists the automatically-created hosts.

2. Verify that the information provided by the HCA is correct (name, host type, host port identifiers).

   If you need to change any of the information, select the host, and then click **View/Edit Settings**.

3. **Optional:** If you want the automatically-created host to be in a cluster, create a host cluster and add the host or hosts.

#### Results

After a host is created automatically, the system displays the following items in the Hosts tile table:

- The host name derived from the system name of the host.
- The host identifier ports that are associated with the host.
- The Host Operating System Type of the host.

## Create host manually

For hosts that cannot be automatically discovered, you can manually create a host. Creating a host is one of the steps required to let the storage array know which hosts are attached to it and to allow I/O access to the volumes.

#### About this task

Keep these guidelines in mind when you create a host:

- You must define the host identifier ports that are associated with the host.
- Make sure that you provide the same name as the host’s assigned system name.
- This operation does not succeed if the name you choose is already in use.
- The length of the name cannot exceed 30 characters.

#### Steps

1. Select **Storage › Hosts**.
2. Click **Create › Host**.

   The **Create Host** dialog box appears.

3. Select the settings for the host as appropriate.

**Field details**

| Setting | Description |
| --- | --- |
| Name | Type a name for the new host. |
| Host operating system type | Select the operating system that is running on the new host from the drop-down list. |
| Host interface type | **Optional:** If you have more than one type of host interface supported on your storage array, select the host interface type that you want to use. |
| Host ports | Do one of the following:<br><br>• **Select I/O Interface**<br><br>Generally, the host ports should have logged in and be available from the drop-down list. You can select the host port identifiers from the list.<br><br>• **Manual add**<br><br>If a host port identifier is not displayed in the list, it means that the host port has not logged in. An HBA utility or the iSCSI initiator utility may be used to find the host port identifiers and associate them with the host.<br><br>You can manually enter the host port identifiers or copy/paste them from the utility (one at a time) into the **Host ports** field.<br><br>You must select one host port identifier at a time to associate it with the host, but you can continue to select as many identifiers that are associated with the host. Each identifier is displayed in the **Host ports** field. If necessary, you also can remove an identifier by selecting the **X** next to it. |

| Setting | Description |
| --- | --- |
| CHAP initiator | **Optional:** If you selected or manually entered a host port with an iSCSI IQN, and if you want to require a host that tries to access the storage array to authenticate using Challenge Handshake Authentication Protocol (CHAP), select the **CHAP initiator** checkbox. For each iSCSI host port you selected or manually entered, do the following:<br>• Enter the same CHAP secret that was set on each iSCSI host initiator for CHAP authentication. If you are using mutual CHAP authentication (two-way authentication that enables a host to validate itself to the storage array and for a storage array to validate itself to the host), you also must set the CHAP secret for the storage array at initial setup or by changing settings.<br>• Leave the field blank if you do not require host authentication. Currently, the only iSCSI authentication method used by System Manager is CHAP. |

4. Click **Create**.

**Results**

After the host is successfully created, the system creates a default name for each host port configured for the host (user label).

The default alias is `<Hostname_Port Number>`. For example, the default alias for the first port created for `host IPT is IPT_1`.

# Create host cluster

## You create a host cluster when two or more hosts require I/O access to the same volumes.

**About this task**

Keep these guidelines in mind when you create a host cluster:

- This operation does not start unless there are two or more hosts available to create the cluster.
- Hosts in host clusters can have different operating systems (heterogeneous).
- To create a Data Assurance (DA)-enabled volume, the host connection you are planning to use must support DA.

  If any of the host connections on the controllers in your storage array do not support DA, the associated hosts cannot access data on DA-enabled volumes.

  DA is **not** supported by iSCSI over TCP/IP, or by the SRP over InfiniBand.

- This operation does not succeed if the name you choose is already in use.
- The length of the name cannot exceed 30 characters.

**Steps**

1. Select **Storage › Hosts**.
2. Select **Create › Host Cluster**.

   The **Create Host Cluster** dialog box appears.

3. Select the settings for the host cluster as appropriate.

   **Field details**

   | Setting | Description |
   |---|---|
   | Name | Type the name for the new host cluster. |
   | Hosts | Select two or more hosts from the drop-down list. Only those hosts that are not already part of a host cluster appear in the list. |

4. Click **Create**.

   If the selected hosts are attached to interface types that have different Data Assurance (DA) capabilities, a dialog box appears with the message that DA will be unavailable on the host cluster. This unavailability prevents DA-enabled volumes from being added to the host cluster. Select **Yes** to continue or **No** to cancel.

   DA increases data integrity across the entire storage system. DA enables the storage array to check for errors that might occur when data is moved between the hosts and the drives. Using DA for the new volume ensures that any errors are detected.

**Results**

The new host cluster appears in the table with the assigned hosts in the rows beneath.

# Create volumes

You create volumes to add storage capacity to an application-specific workload, and to make the created volumes visible to a specific host or host cluster. In addition, the volume creation sequence provides options to allocate specific amounts of capacity to each volume you want to create.

**About this task**

Most application types default to a user-defined volume configuration. Some application types have a smart configuration applied at volume creation. For example, if you are creating volumes for Microsoft Exchange application, you are asked how many mailboxes you need, what your average mailbox capacity requirements are, and how many copies of the database you want. System Manager uses this information to create an optimal volume configuration for you, which can be edited as needed.

> If you want to mirror a volume, first create the volumes that you want to mirror, and then use the **Storage › Volumes › Copy Services › Mirror a volume asynchronously** option.

The process to create a volume is a multi-step procedure.

## Step 1: Select host for a volume

You create volumes to add storage capacity to an application-specific workload, and to make the created volumes visible to a specific host or host cluster. In addition, the volume creation sequence provides options to allocate specific amounts of capacity to each volume you want to create.

### Before you begin

- Valid hosts or host clusters exist under the **Hosts** tile.
- Host port identifiers have been defined for the host.
- Before creating a DA-enabled volume, the host connection you are planning to use must support DA. If any of the host connections on the controllers in your storage array do not support DA, the associated hosts cannot access data on DA-enabled volumes.

### About this task

Keep these guidelines in mind when you assign volumes:

- A host's operating system can have specific limits on how many volumes the host can access. Keep this limitation in mind when you create volumes for use by a particular host.
- You can define one assignment for each volume in the storage array.
- Assigned volumes are shared between controllers in the storage array.
- The same logical unit number (LUN) cannot be used twice by a host or a host cluster to access a volume. You must use a unique LUN.

> Assigning a volume to a host will fail if you try to assign a volume to a host cluster that conflicts with an established assignment for a host in the host clusters.

### Steps

1. Select **Storage › Volumes**.
2. Select **Create › Volume**.

   The **Create Volumes** dialog box appears.

3. From the drop-down list, select a specific host or host cluster to which you want to assign volumes, or choose to assign the host or host cluster at a later time.
4. To continue the volume creation sequence for the selected host or host cluster, click **Next**, and go to Step 2: Select a workload for a volume.

   The **Select Workload** dialog box appears.

## Step 2: Select a workload for a volume

Select a workload to customize the storage array configuration for a specific application, such as Microsoft SQL Server, Microsoft Exchange, Video Surveillance applications, or VMware. You can select "Other

application" if the application you intend to use on this storage array is not listed.

#### About this task

This task describes how to create volumes for an existing workload.

- *When you are creating volumes using an application-specific workload*, the system may recommend an optimized volume configuration to minimize contention between application workload I/O and other traffic from your application instance. You can review the recommended volume configuration and edit, add, or delete the system-recommended volumes and characteristics using the **Add/Edit Volumes** dialog box.
- *When you are creating volumes using "Other"* applications (or applications without specific volume creation support), you manually specify the volume configuration using the **Add/Edit Volumes** dialog box.

#### Steps

1. Do one of the following:
    - Select the **Create volumes for an existing workload** option to create volumes for an existing workload.
    - Select the **Create a new workload** option to define a new workload for a supported application or for "Other" applications.
        - From the drop-down list, select the name of the application you want to create the new workload for.

          Select one of the "Other" entries if the application you intend to use on this storage array is not listed.
        - Enter a name for the workload you want to create.
2. Click **Next**.
3. If your workload is associated with a supported application type, enter the information requested; otherwise, go to Step 3: Add or edit volumes.

## Step 3: Add or edit volumes

#### Before you begin

- The pools or volume groups must have sufficient free capacity.
- To create a Data Assurance (DA)-enabled volume, the host connection you are planning to use must support DA.

**Selecting a DA capable pool or volume group**

If you want to create a DA-enabled volume, select a pool or volume group that is DA capable (look for **Yes** next to "DA" in the pool and volume group candidates table).

DA capabilities are presented at the pool and volume group level in System Manager. DA protection checks for and corrects errors that might occur as data is transferred through the controllers down to the drives. Selecting a DA-capable pool or volume group for the new volume ensures that any errors are detected and corrected.

If any of the host connections on the controllers in your storage array do not support DA, the associated hosts cannot access data on DA-enabled volumes.

DA is not supported by iSCSI over TCP/IP, or by the SRP over InfiniBand.

- To create a secure-enabled volume, a security key must be created for the storage array.

**Selecting a secure-capable pool or volume group**

If you want to create a secure-enabled volume, select a pool or volume group that is secure capable (look for **Yes** next to "Secure-capable" in the pool and volume group candidates table).

Drive security capabilities are presented at the pool and volume group level in System Manager. Secure-capable drives prevent unauthorized access to the data on a drive that is physically removed from the storage array. A secure-enabled drive encrypts data during writes and decrypts data during reads using a unique *encryption key*.

A pool or volume group can contain both secure-capable and non-secure-capable drives, but all drives must be secure-capable to use their encryption capabilities.

**About this task**

You create volumes from pools or volume groups. The **Add/Edit Volumes** dialog box shows all eligible pools and volume groups on the storage array. For each eligible pool and volume group, the number of drives available and the total free capacity appears.

For some application-specific workloads, each eligible pool or volume group shows the proposed capacity based on the suggested volume configuration and shows the remaining free capacity in GiB. For other workloads, the proposed capacity appears as you add volumes to a pool or volume group and specify the reported capacity.

**Steps**

1. Choose one of these actions based on whether you selected Other or an application-specific workload:
   - **Other** — Click **Add new volume** in each pool or volume group that you want to use to create one or more volumes.

**Field Details**

| Field | Description |
| --- | --- |
| Volume Name | A volume is assigned a default name by System Manager during the volume creation sequence. You can either accept the default name or provide a more descriptive one indicating the type of data stored in the volume. |
| Reported Capacity | Define the capacity of the new volume and the capacity units to use (MiB, GiB, or TiB). For **Thick volumes**, the minimum capacity is 1 MiB, and the maximum capacity is determined by the number and capacity of the drives in the pool or volume group.<br><br>Keep in mind that storage capacity is also required for copy services (snapshot images, snapshot volumes, volume copies, and remote mirrors); therefore, do not allocate all of the capacity to standard volumes.<br><br>Capacity in a pool is allocated in 4-GiB increments. Any capacity that is not a multiple of 4 GiB is allocated but not usable. To make sure that the entire capacity is usable, specify the capacity in 4-GiB increments. If unusable capacity exists, the only way to regain it is to increase the capacity of the volume. |

| Field | Description |
| --- | --- |
| Segment Size | Shows the setting for segment sizing, which only appears for volumes in a volume group. You can change the segment size to optimize performance.<br><br>**Allowed segment size transitions** — System Manager determines the segment size transitions that are allowed. Segment sizes that are inappropriate transitions from the current segment size are unavailable on the drop-down list. Allowed transitions usually are double or half of the current segment size. For example, if the current volume segment size is 32 KiB, a new volume segment size of either 16 KiB or 64 KiB is allowed.<br><br>**SSD Cache-enabled volumes** — You can specify a 4-KiB segment size for SSD Cache-enabled volumes. Make sure you select the 4-KiB segment size only for SSD Cache-enabled volumes that handle small-block I/O operations (for example, 16 KiB I/O block sizes or smaller). Performance might be impacted if you select 4 KiB as the segment size for SSD Cache-enabled volumes that handle large block sequential operations.<br><br>**Amount of time to change segment size** — The amount of time to change a volume's segment size depends on these variables:<br><br>• The I/O load from the host<br>• The modification priority of the volume<br>• The number of drives in the volume group<br>• The number of drive channels<br>• The processing power of the storage array controllers<br><br>When you change the segment size for a volume, I/O performance is affected, but your data remains available. |

| Field | Description |
| --- | --- |
| Secure-capable | **Yes** appears next to "Secure-capable" only if the drives in the pool or volume group are secure-capable.<br><br>Drive Security prevents unauthorized access to the data on a drive that is physically removed from the storage array. This option is available only when the Drive Security feature has been enabled, and a security key is set up for the storage array.<br><br>A pool or volume group can contain both secure-capable and non-secure-capable drives, but all drives must be secure-capable to use their encryption capabilities. |
| DA | **Yes** appears next to "DA" only if the drives in the pool or volume group support Data Assurance (DA).<br><br>DA increases data integrity across the entire storage system. DA enables the storage array to check for errors that might occur as data is transferred through the controllers down to the drives. Using DA for the new volume ensures that any errors are detected. |

- **Application-specific workload** — Either click **Next** to accept the system-recommended volumes and characteristics for the selected workload, or click **Edit Volumes** to change, add, or delete the system-recommended volumes and characteristics for the selected workload.

**Field Details**

| Field | Description |
| --- | --- |
| Volume Name | A volume is assigned a default name by System Manager during the volume creation sequence. You can either accept the default name or provide a more descriptive one indicating the type of data stored in the volume. |
| Reported Capacity | Define the capacity of the new volume and the capacity units to use (MiB, GiB, or TiB). For **Thick volumes**, the minimum capacity is 1 MiB, and the maximum capacity is determined by the number and capacity of the drives in the pool or volume group.<br><br>Keep in mind that storage capacity is also required for copy services (snapshot images, snapshot volumes, volume copies, and remote mirrors); therefore, do not allocate all of the capacity to standard volumes.<br><br>Capacity in a pool is allocated in 4-GiB increments. Any capacity that is not a multiple of 4 GiB is allocated but not usable. To make sure that the entire capacity is usable, specify the capacity in 4-GiB increments. If unusable capacity exists, the only way to regain it is to increase the capacity of the volume. |
| Volume Type | Volume type indicates the type of volume that was created for an application-specific workload. |

| Field | Description |
| --- | --- |
| Segment Size | Shows the setting for segment sizing, which only appears for volumes in a volume group. You can change the segment size to optimize performance.<br><br>**Allowed segment size transitions** — System Manager determines the segment size transitions that are allowed. Segment sizes that are inappropriate transitions from the current segment size are unavailable on the drop-down list. Allowed transitions usually are double or half of the current segment size. For example, if the current volume segment size is 32 KiB, a new volume segment size of either 16 KiB or 64 KiB is allowed.<br><br>**SSD Cache-enabled volumes** — You can specify a 4-KiB segment size for SSD Cache-enabled volumes. Make sure you select the 4-KiB segment size only for SSD Cache-enabled volumes that handle small-block I/O operations (for example, 16 KiB I/O block sizes or smaller). Performance might be impacted if you select 4 KiB as the segment size for SSD Cache-enabled volumes that handle large block sequential operations.<br><br>**Amount of time to change segment size** — The amount of time to change a volume's segment size depends on these variables:<br><br>• The I/O load from the host<br>• The modification priority of the volume<br>• The number of drives in the volume group<br>• The number of drive channels<br>• The processing power of the storage array controllers<br><br>When you change the segment size for a volume, I/O performance is affected, but your data remains available. |
| Secure-capable | **Yes** appears next to "Secure-capable" only if the drives in the pool or volume group are secure-capable.<br><br>Drive security prevents unauthorized access to the data on a drive that is physically removed from the storage array. This option is available only when the drive security feature has been enabled, and a security key is set up for the storage array.<br><br>A pool or volume group can contain both secure-capable and non-secure-capable drives, but all drives must be secure-capable to use their encryption capabilities. |

| Field | Description |
| --- | --- |
| DA | **Yes** appears next to "DA" only if the drives in the pool or volume group support Data Assurance (DA).<br><br>DA increases data integrity across the entire storage system. DA enables the storage array to check for errors that might occur as data is transferred through the controllers down to the drives. Using DA for the new volume ensures that any errors are detected. |

2. To continue the volume creation sequence for the selected application, click **Next**, and go to Step 4: Review volume configuration.

### Step 4: Review volume configuration

Review a summary of the volumes you intend to create and make any necessary changes.

#### Steps

1. Review the volumes you want to create. Click **Back** to make any changes.
2. When you are satisfied with your volume configuration, click **Finish**.

#### Results

System Manager creates the new volumes in the selected pools and volume groups, and then displays the new volumes in the All Volumes table.

#### After you finish

- Perform any operating system modifications necessary on the application host so that the applications can use the volume.
- Run either the host-based `hot_add` utility or an operating system-specific utility (available from a third-party vendor), and then run the `SMdevices` utility to correlate volume names with host storage array names.

  The `hot_add` utility and the `SMdevices` utility are included as part of the `SMutils` package. The `SMutils` package is a collection of utilities to verify what the host sees from the storage array. It is included as part of the SANtricity software installation.

## Assign volumes

You must assign a volume to a host or a host cluster so it can be used for I/O operations. This assignment grants a host or host cluster access to one or more volumes in a storage array.

#### Before you begin

Keep these guidelines in mind when you assign volumes:

- You can assign a volume to only one host or host cluster at a time.
- Assigned volumes are shared between controllers in the storage array.
- The same logical unit number (LUN) cannot be used twice by a host or a host cluster to access a volume.

You must use a unique LUN.

Assigning a volume fails under these conditions:

- All volumes are assigned.
- The volume is already assigned to another host or host cluster.

The ability to assign a volume is unavailable under these conditions:

- No valid hosts or host clusters exist.
- No host port identifiers have been defined for the host.
- All volume assignments have been defined.

**About this task**

All unassigned volumes are displayed, but functions for hosts with or without Data Assurance (DA) apply as follows:

- For a DA-capable host, you can select volumes that are either DA-enabled or not DA-enabled.
- For a host that is not DA-capable, if you select a volume that is DA-enabled, a warning states that the system must automatically turn off DA on the volume before assigning the volume to the host.

**Steps**

1. Select **Storage › Hosts**.
2. Select the host or host cluster to which you want to assign volumes, and then click **Assign Volumes**.

   A dialog box appears that lists all the volumes that can be assigned. You can sort any of the columns or type something in the **Filter** box to make it easier to find particular volumes.

3. Select the check box next to each volume that you want to assign or select the check box in the table header to select all volumes.
4. Click **Assign** to complete the operation.

**Results**

After successfully assigning a volume or volumes to a host or a host cluster, the system performs the following actions:

- The assigned volume receives the next available LUN number. The host uses the LUN number to access the volume.
- The user-supplied volume name appears in volume listings associated to the host. If applicable, the factory-configured access volume also appears in volume listings associated to the host.

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